

The Impact of Armed Conflict on Commodity Prices and Analysis (2019–2025)

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Motivation

There has always been a certain connection between armed conflicts and the fluctuations in commodity prices: Russia's invasion of Ukraine in 2022 led to an immediate surge in crude oil and natural gas prices, while the Gaza conflict in 2023 pushed gold prices to a record high as investors flocked to safe-haven assets. However, a single conflict merely constitutes an observation point, and thus the conclusion drawn is prone to be influenced by confounding factors such as adjustments in monetary policy or disruptions in the supply chain during the same period. This study takes three conflicts, namely Russia-Ukraine, Israel-Palestine and Yemen, which are geographically and structurally distinct, as independent test cases, aiming to answer the following question: Can the intensity of conflicts continuously predict the trends of energy and gold prices in different regions and asset classes? If so, what about its time lag effect? The utilization of multiple conflict cases can not only significantly enhance the credibility of the observed patterns, but also reveal which bulk commodities are most sensitive to what types of conflicts.

Data Sources

#	Source	Extraction Method	Data / URL
1 (hardest)	ACLED – Armed Conflict Location & Event Data	REST API via requests; pagination loop (500-row limit per call); downloads ALL global events; region filtering in clean_data.py	https://acleddata.com/ https://api.acleddata.com/acleddata/read Raw: all global events (2019–2025, ~1M rows) Fields: date, event_type, fatalities
2	U.S. Energy Information Administration (EIA)	REST API via requests; multi-series loop; free API key required	https://www.eia.gov/opendata/ https://api.eia.gov/v2/ Series: WTI crude, Brent crude, Henry Hub gas, heating oil, coal (daily & monthly prices)
3	Yahoo Finance – Gold Futures (GC=F)	yfinance Python library (pip install yfinance) no authentication needed	https://finance.yahoo.com/quote/GC=F/ yf.download('GC=F', start='2019-01-01', end='2025-12-31') Daily OHLCV gold futures price

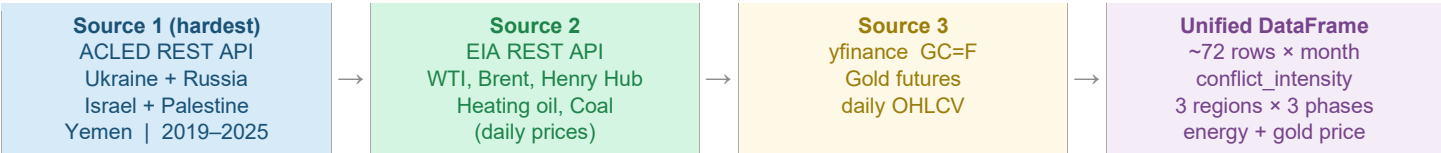


Figure 1. Data integration schema: three sources joined on year-month key; ACLED filtered to three conflict regions serving as independent test cases.

Data Integration and Research Questions

Raw ACLED data is first cleaned in clean_data.py: (1) filter to five countries (Ukraine, Russia, Israel, Palestine, Yemen); (2) retain battle/violence event types only; (3) impute missing fatalities with 0; (4) standardize to year-month format; (5) deduplicate on event_id — reducing ~1M rows to ~80 – 120K rows aggregated monthly. Merge these three cleaned sources by year and month keys to form a unified DataFrame of approximately 72 lines for the following three analyses:

- (1) Global correlation: Pearson correlation coefficient r with various commodities in the 0 to 8-week lag period;
- (2) Comparison between conflicts: Has the Russia-Ukraine conflict driven up energy prices, while have the Israel-Palestine and Yemen conflicts driven up gold prices?
- (3) Anomaly Detection in case Studies: Comparison of price anomalies with the baseline during the escalation of the situation in February 2022 and October 2023.

This analysis aims to establish correlations rather than causal relationships. Macroeconomic confounding factors cannot be completely ruled out.

References: [1] ACLED. (2025). Armed Conflict Location & Event Data. <https://acleddata.com/> [2] U.S. Energy Information Administration. (2025). EIA Open Data API v2. <https://www.eia.gov/opendata/> [3] Aroussi, R. (2023). yfinance: Yahoo! Finance market data downloader. <https://github.com/ranaroussi/yfinance>